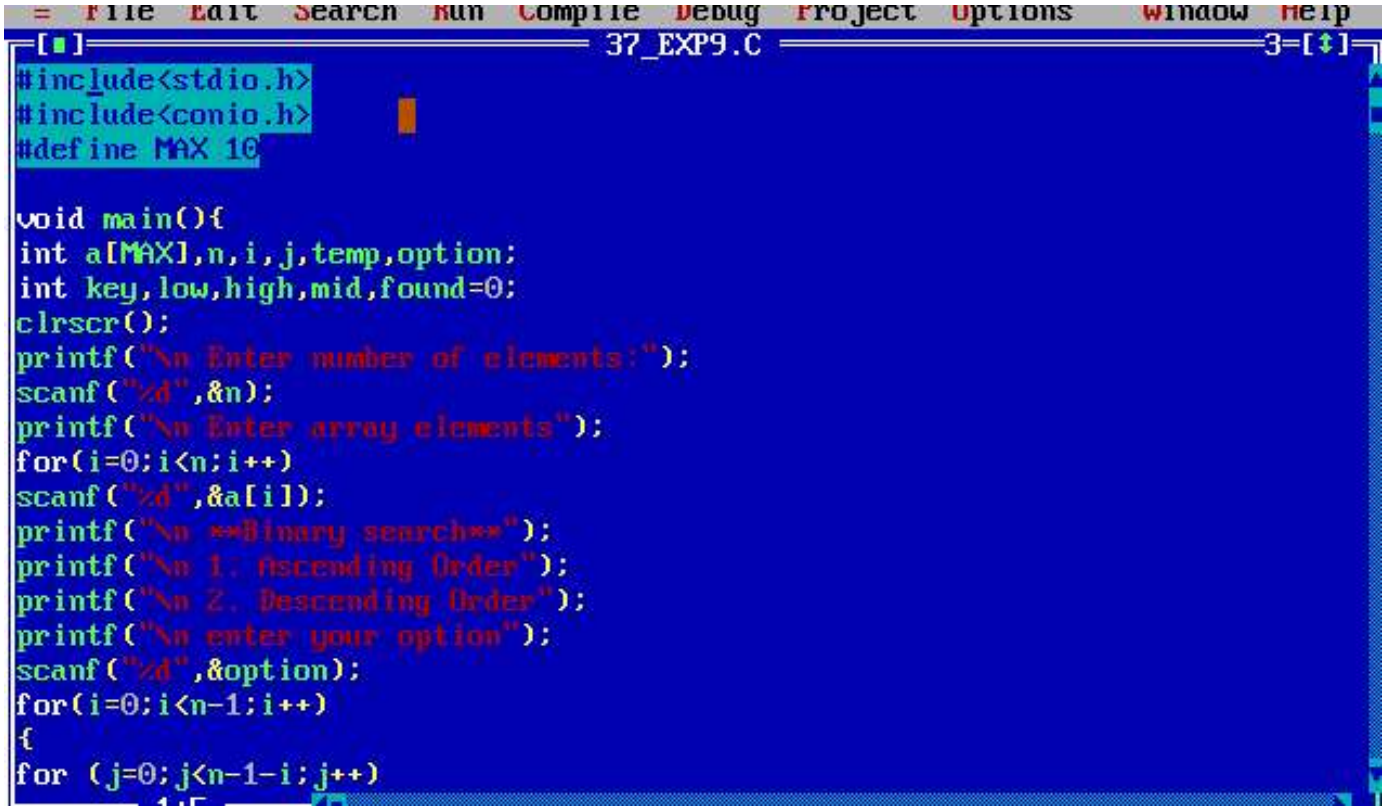
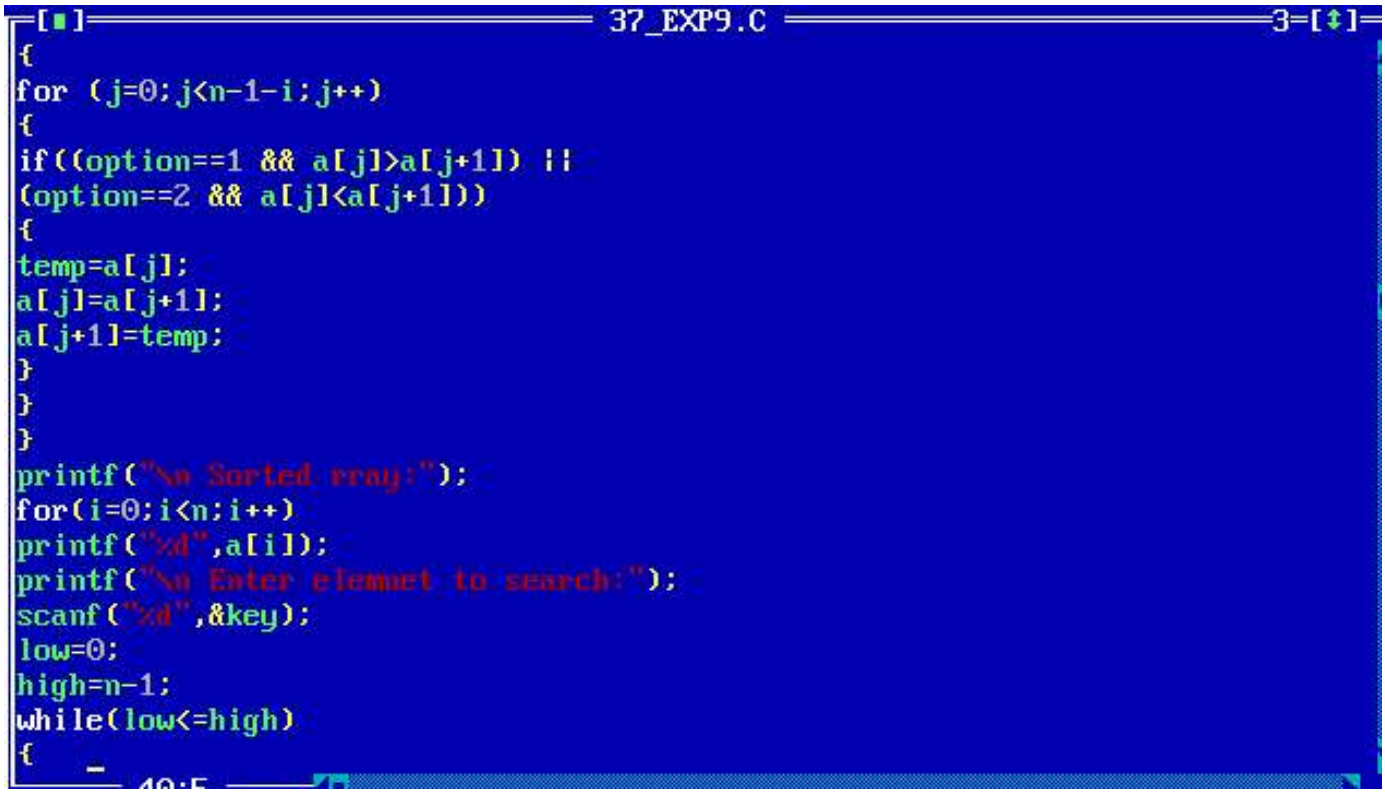
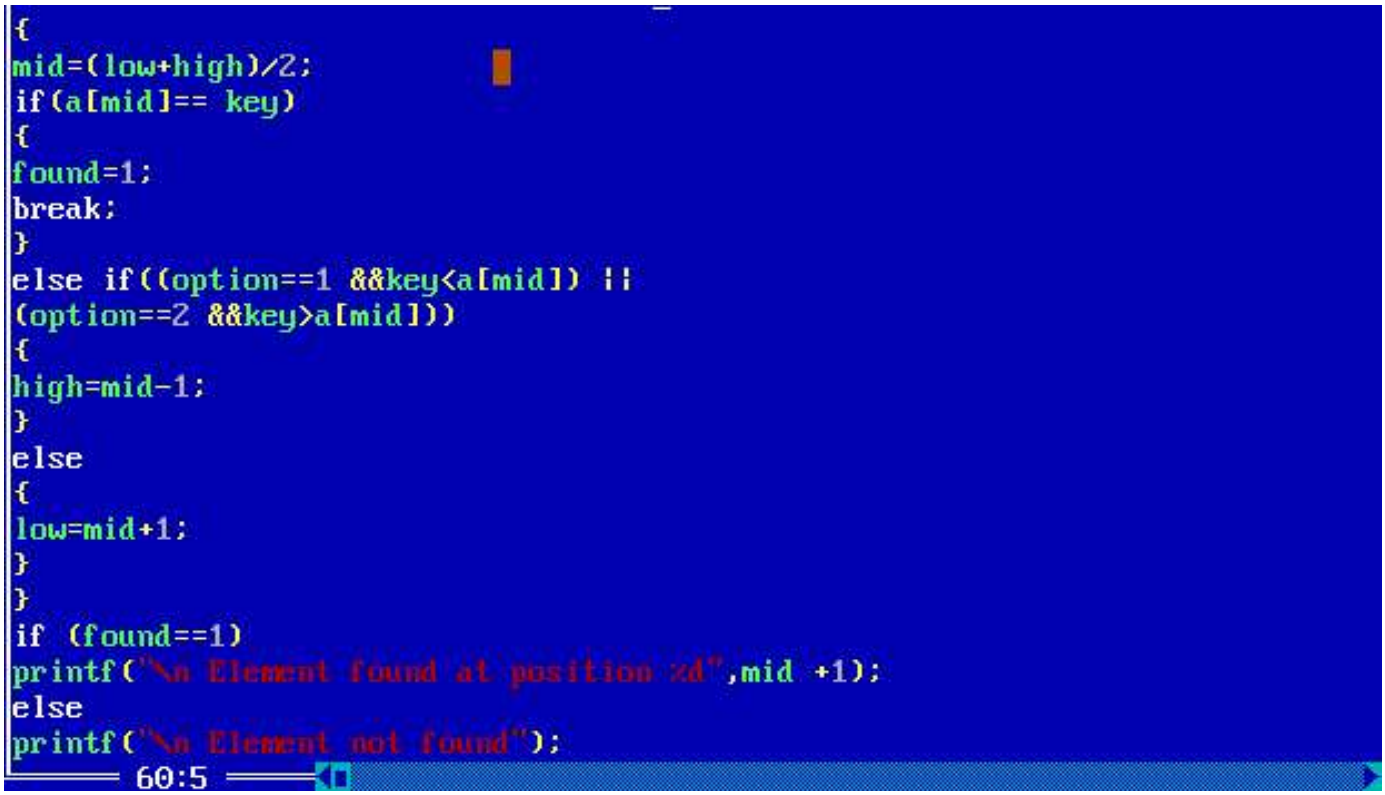
Code (part 1)


File Edit Search Run Compile Debug Project Options Window Help
37_EXP9.C
#include<stdio.h>
#include<conio.h>
#define MAX 10
void main(){
int a[MAX],n,i,j,temp,option;
int key,low,high,mid,found=0;
clrscr();
printf("\n Enter number of elements:");
scanf("%d",&n);
printf("\n Enter array elements");
for(i=0;i<n;i++)
scanf("%d",&a[i]);
printf("\n **Binary search**");
printf("\n 1. Ascending Order");
printf("\n 2. Descending Order");
printf("\n enter your option");
scanf("%d",&option);
for(i=0;i<n-1;i++)
{
for (j=0;j<n-1-i;j++)

Code (part 2)

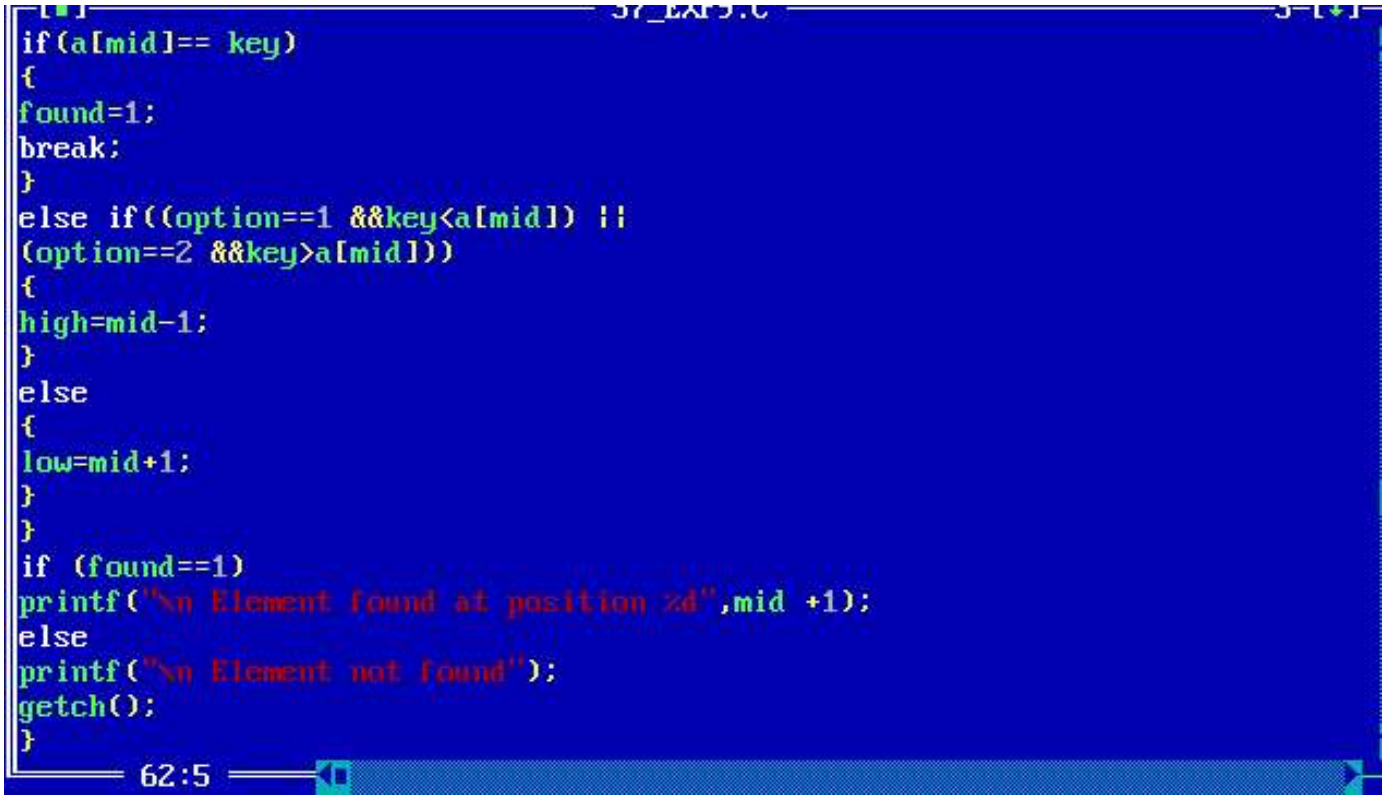

37_EXP9.C
{
for (j=0;j<n-1-i;j++)
{
if((option==1 && a[j]>a[j+1]) ||
(option==2 && a[j]<a[j+1]))
{
temp=a[j];
a[j]=a[j+1];
a[j+1]=temp;
}
}
}
printf("\n Sorted rray:");
for(i=0;i<n;i++)
printf("%d",a[i]);
printf("\n Enter elemnet to search:");
scanf("%d",&key);
low=0;
high=n-1;
while(low<=high)
{

Code (part 3)



```
{
mid=(low+high)/2;
if(a[mid]== key)
{
found=1;
break;
}
else if((option==1 &&key<a[mid]) ||
(option==2 &&key>a[mid]))
{
high=mid-1;
}
else
{
low=mid+1;
}
}
if (found==1)
printf("\n Element found at position %d",mid +1);
else
printf("\n Element not found");
```

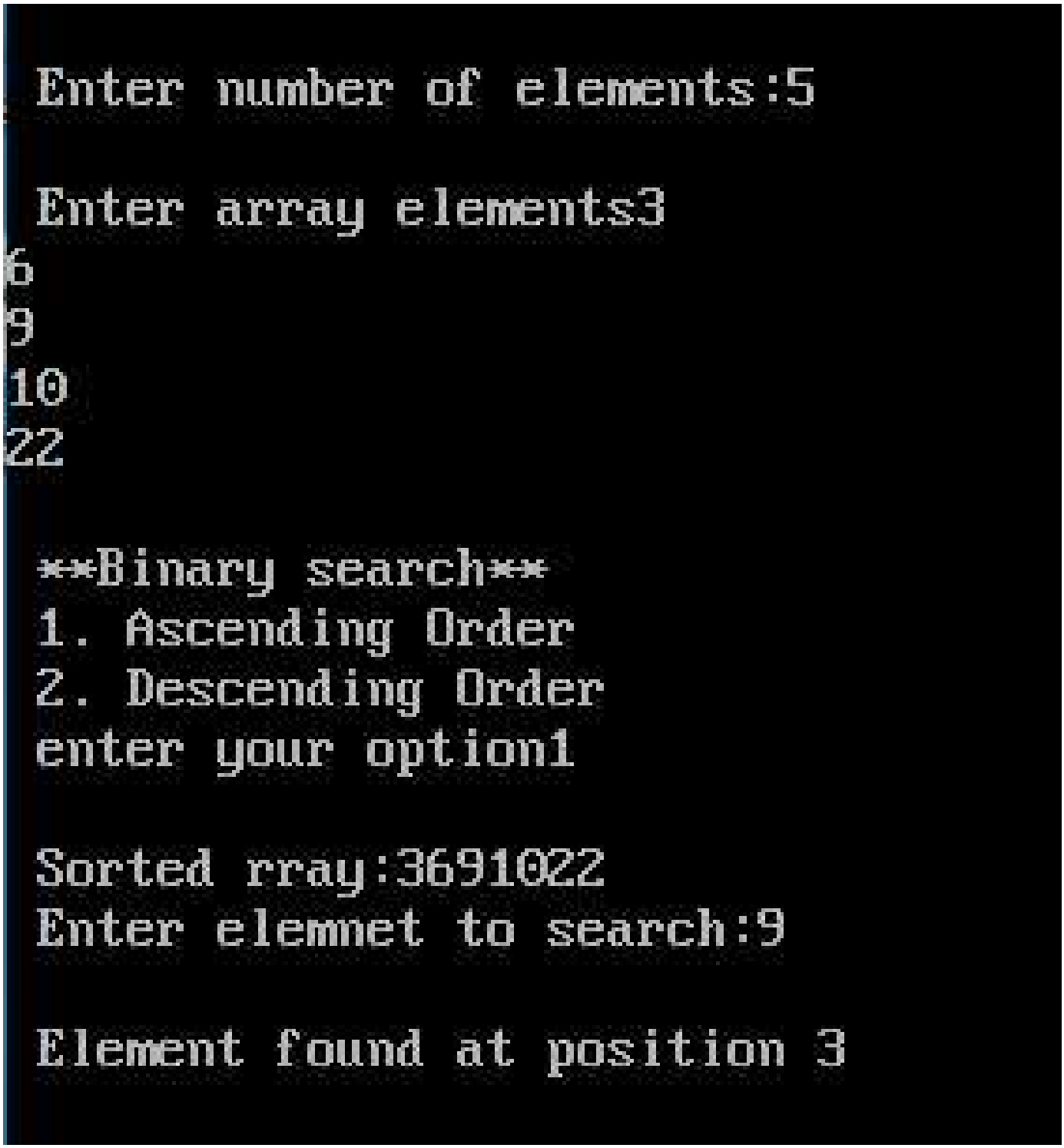
Code (part 4)



```
if(a[mid]== key)
{
found=1;
break;
}
else if((option==1 &&key<a[mid]) ||
(option==2 &&key>a[mid]))
{
high=mid-1;
}
else
{
low=mid+1;
}
}
if (found==1)
printf("\n Element found at position %d",mid +1);
else
printf("\n Element not found");
getch();
}
```

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Output


Enter number of elements:5
Enter array elements3
6
9
10
22
Binary search
1. Ascending Order
2. Descending Order
enter your option1
Sorted rray:3691022
Enter elemnet to search:9
Element found at position 3